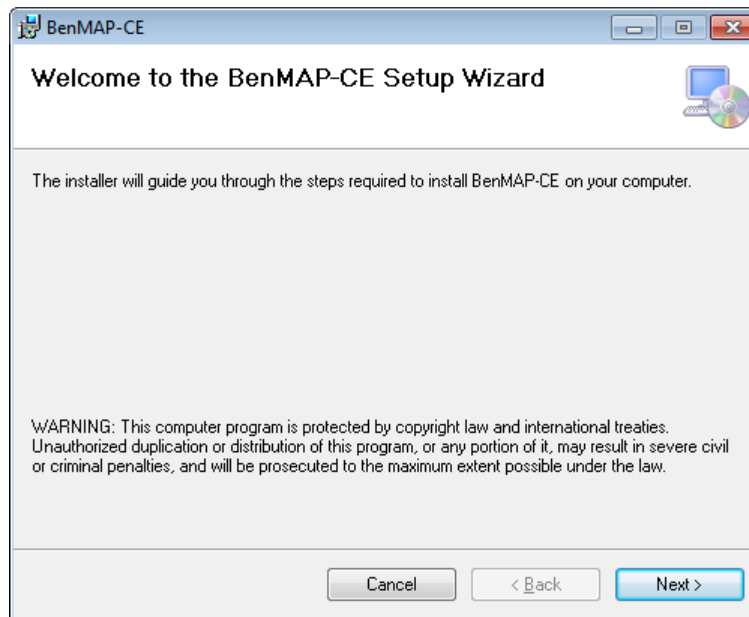


- At least 4 gigabytes of RAM⁷
- Intel or compatible processor, Core i5 (or better)
- At least 10 GB free disk space is necessary for installation of the BenMAP-CE database and ancillary files⁸.

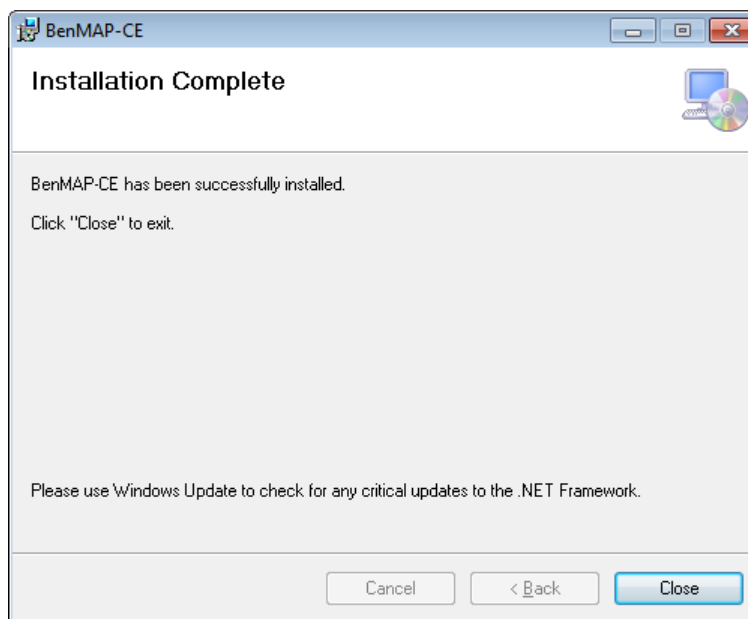
1.4 Installing BenMAP-CE

The installation of BenMAP-CE is very simple. Double click **Setup.exe** in your installation directory to bring up the setup wizard. Then follow the setup wizard by clicking 'Next' or 'OK' to complete the installation.



⁷ BenMAP-CE works best in a 64-bit Windows environment. With a 32-bit installation there are limits on the memory available to the software application; it can utilize no more than 2 GB of RAM. This will impact performance when processing large spatial datasets or numerous health impact/valuation functions. To determine whether your computer is running a 32-bit or 64-bit version of Windows, refer to this article from Microsoft: <https://support.microsoft.com/en-us/help/15056/windows-7-32-64-bit-faq>

⁸ A solid state drive (SSD) has also shown improved performance over hard disk drives (HDD).



Upgrading to a new version of BenMAP-CE

Periodically, new versions of BenMAP-CE will be made available and posted to the BenMAP-CE website: <http://www.epa.gov/benmap/>. If you are upgrading, first uninstall the previous version of the software on your computer (details provided in Section 1.5). Then extract the files from the installer package (.zip file) and run the executable ("setup.exe"). A new installer package can be relatively large (~1.2 gigabytes) because the database is embedded in the installer. However, once downloaded, the installation process is generally very fast.

Installation instructions are typically provided with each software release. Refer to these supplemental instructions for important additional information. Please note that your existing Setups will not be automatically transferred to newly installed versions of BenMAP. See instructions for exporting and importing databases in Chapter 9.

1.5 Uninstalling BenMAP-CE

To uninstall BenMAP-CE, go to **Control Panel, Programs and Features** and remove **BenMAP-CE**. Note that uninstalling BenMAP-CE does not also remove any results files that you have created with BenMAP-CE.

1.6 Contacts for Comments, Questions & Bug Reporting

For comments and questions, please contact Peter Maniloff at the U.S. EPA.

Address: C539-07, U.S. EPA Mailroom, Research Triangle Park, NC 27711

Email: maniloff.peter@epa.gov

Telephone: 919-541-5548

Alternatively, you can send a message at the BenMAP-CE website:

<https://www.epa.gov/benmap/forms/contact-us-about-benmap>, or by simply emailing benmap@epa.gov.

To report programming bugs or suggest additions to the software in BenMAP-CE:

- Select the **Help** menu in the main window;
- Open the **Provide Feedback** form;
- Complete the form and submit the report.

Provide Feedback

BenMAP-CE is open-source software. If you encounter an error or would like to request a feature, please consider submitting a report by filling out this form. We apologize for any issues you may have encountered and value your feedback as it will help us continue to improve BenMAP-CE.

Operating System: Microsoft Windows NT 6.2.9200.0

BenMAP-CE Version: 1.4.8.0 (June 12, 2018)

Your name (optional):

Your email address (optional):

Country where you are located (optional):

What type of report would you like to submit?

☒ Software Error

☐ Requested Feature

How severe is the error you experienced?

☐ Minor - this issue has little or no impact on my ability to use BenMAP-CE

☒ Major - this issue significantly hinders my ability to use BenMAP-CE

☐ Blocking - this issue prevents me from using BenMAP-CE

What component of BenMAP-CE does the error affect?

Graphic User Interface: suggested changes

The audit trail report may help us to debug or understand your report better.

☒ Include BenMAP-CE generated audit trail report.

Please describe what you were doing when you encountered the error. Can you tell us how to reproduce the error? (5000 character limit)

To submit this report to us, click the Submit button. Thank you for your time.

Submit Cancel

1.7 Sources for More Information

For supplemental information on BenMAP-CE, such as articles and presentations, manuals, and training materials go to:

- U.S. Environmental Protection Agency, Office of Air Quality Planning and Standards (OAQPS), BenMAP-CE website: <https://www.epa.gov/benmap/>

To interact with other BenMAP users, go to:

- BenMAP-CE Discussion Forum: <https://forum.benmap.org/>

To access the BenMAP-CE source code, go to:

- BenMAP-CE on Github: <https://github.com/BenMAPCE/BenMAP-CE>

For more information on conducting benefit analysis, see the following documents:

- U.S. EPA (various years). Costs and Benefits of the Clean Air Act. Available at: <https://www.epa.gov/clean-air-act-overview/benefits-and-costs-clean-air-act>
- U.S. EPA (2006). Final Regulatory Impact Analysis: 2006 National Ambient Air Quality Standards for Particulate Matter. Office of Air Quality Planning and Standards. *See:* Chapter 5. Available at: <http://www.epa.gov/ttn/ecas/regdata/RIAs/Chapter%205--Benefits.pdf>
- U.S. EPA (2008). Final Ozone NAAQS Regulatory Impact Analysis. Office of Air Quality Planning and Standards. March. *See:* Chapter 6. Available at: <https://www3.epa.gov/ttn/ecas/regdata/RIAs/6-ozoneriachapter6.pdf>
- U.S. EPA (2008). Regulatory Impact Analysis: Control of Emissions of Air Pollution from Locomotive Engines and Marine Compression Ignition Engines Less than 30 Liters Per Cylinder. Office of Transportation and Air Quality. EPA420-R-08-001a. May. *See:* Chapter 6. Available at: <https://nepis.epa.gov/Exe/ZyPDF.cgi/P10024CN.PDF?Dockey=P10024CN.PDF>
- U.S. EPA (2010). Final Regulatory Impact Analysis (RIA) for the SO₂ National Ambient Air Quality Standards (NAAQS). Office of Air Quality Planning and Standards. June. *See:* Chapter 5. Available at: <http://www.epa.gov/ttn/ecas/regdata/RIAs/fso2ria100602ch5.pdf>
- U.S. EPA (2010). Final Regulatory Impact Analysis (RIA) for the NO₂ National Ambient Air Quality Standards (NAAQS). Office of Air Quality Planning and Standards. June. *See:* Chapter 4. Available at: https://www3.epa.gov/ttn/ecas/docs/ria/naaqs-no2_ria_final_2010-01.pdf